CYBERSECURITY TRAINING

Curriculum

Practical-first • Beginner-friendly • Lab-based • Career-oriented

September | 2026

1. Programme Overview

Course purpose

This curriculum is designed to give complete beginners a structured entry into cybersecurity. Learners first build essential computer, operating system and networking knowledge, then progress into identity protection, endpoint and network defense, security testing, vulnerability management, web security, monitoring, incident response, digital forensics, governance and a final integrated capstone. The course emphasizes responsible, authorized and defensive use of cybersecurity tools.

Target participants

- Young people and graduates seeking an entry point into cybersecurity.
- IT support staff who need foundational cybersecurity skills.
- Students or career switchers with basic computer literacy.
- Early-career technology professionals who need practical security exposure.

Entry requirements

- Basic ability to use a computer, browser, email and common office applications.
- Willingness to work with Windows and Linux virtual machines.
- No coding background is required.
- Learners must agree to the course ethics and authorized-lab-use rules before security testing activities.

Programme learning outcomes

- Explain core cybersecurity concepts, threats, vulnerabilities, risk and security controls.
- Use Windows, Linux and networking fundamentals to investigate basic security issues.
- Apply identity, endpoint, network, wireless, cloud and data-protection controls.
- Capture and interpret basic network traffic and security logs.
- Perform authorized host/service discovery and interpret vulnerability information within a controlled lab.
- Recognize common web application security weaknesses and explain appropriate remediation.
- Participate in basic security monitoring, incident response and digital evidence handling.
- Create simple risk registers, security policies, incident reports and remediation recommendations.
- Communicate cybersecurity findings clearly to technical and non-technical audiences.
- Demonstrate integrated beginner-level cybersecurity skills through a practical capstone.

2. Curriculum Architecture

Four progressive learning phases across 20 training days

Phase	Days	Focus	Key learner outcome
Week 1: Foundations	1-5	Cybersecurity concepts, computers, networking, Linux and cyber threats.	Understand how systems work and how common cyber threats exploit people and technology.
Week 2: Protection & Defense	6-10	Identity, Windows security, network defense, packet analysis and ethical security testing.	Apply core protective controls and use basic defensive/security-testing tools safely.
Week 3: Detection & Application Security	11-15	Vulnerabilities, web security, cryptography, cloud security and monitoring.	Identify weaknesses, protect data and analyze security evidence.
Week 4: Response, Governance & Career Readiness	16-20	Incident response, forensics, governance, security operations and capstone.	Respond to incidents, document risk and demonstrate integrated beginner cybersecurity skills.

Standard five-hour learning model

Time	Learning Block	Focus / Activity
45 min	Foundation	Recap, prior-knowledge activation, concepts and learning objectives.
75 min	Core Instruction	Main concepts, examples, instructor explanation and guided questioning.
45 min	Guided Demonstration	Trainer demonstrates relevant tools, workflows or security controls.
105 min	Hands-on Practice	Individual or team lab, scenario, investigation or applied exercise.
30 min	Review & Assessment	Quiz, practical check, peer review, reflection or challenge debrief.

Note: Short comfort breaks may be scheduled by the facilitator without changing the learning objectives. The institution may extend the calendar day where breaks must sit outside the five contact hours.

3. Detailed 20-Day Curriculum

Daily objectives, core content, practical work and five-hour delivery plans

DAY 1

Cybersecurity Foundations & the Modern Threat Landscape | Week 1 - Foundations | 5 hours

Learning objectives: By the end of Day 1, learners should be able to:

- Explain cybersecurity in practical terms and why it matters to individuals, businesses and public institutions.
- Distinguish assets, threats, vulnerabilities, risk and controls.
- Explain confidentiality, integrity and availability (CIA) using everyday examples.
- Recognize common cyber incidents and responsible cybersecurity behaviour.

Core content

- Cybersecurity concepts and terminology; digital assets and attack surfaces.
- CIA triad, authentication, authorization, accountability and non-repudiation.
- Common threat actors and motivations: cybercriminals, insiders, hacktivists and state-linked actors.
- Overview of phishing, malware, ransomware, credential theft, fraud and data breaches.
- Cybersecurity ethics, authorization, safe lab rules and responsible disclosure.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Course orientation, baseline knowledge check and discussion of everyday cyber risks.
75 min	Core Instruction	Cybersecurity principles, assets, threats, vulnerabilities, risks and controls.
45 min	Guided Demonstration	Walk through a fictional breach and map the failure points to CIA and basic controls.
105 min	Hands-on Practice	Team exercise: create an asset-and-risk map for a small organization and recommend basic safeguards.
30 min	Review & Assessment	10-question quiz, review of terminology and learner reflection.

Practical lab / learner output: Completed organizational asset-and-risk map with at least five assets, associated threats, vulnerabilities and recommended controls.

Assessment: Baseline diagnostic (ungraded) plus Day 1 knowledge quiz.

DAY 2

Computer, Operating System & File Security Fundamentals | Week 1 - Foundations | 5 hours

Learning objectives: By the end of Day 2, learners should be able to:

- Explain the basic components of a computer and how an operating system manages them.
- Navigate files, users, processes and services in Windows and Linux.
- Understand file permissions, user privileges and why administrative access is sensitive.
- Apply basic secure-computing practices to a workstation.

Core content

- CPU, memory, storage, processes, services, applications and file systems.
- Windows and Linux user accounts, administrative/root privileges and system services.
- File paths, extensions, hidden files, permissions and ownership.
- Software installation, patching, updates and application trust.
- Basic endpoint hardening concepts.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Recap and computer architecture for cybersecurity.
75 min	Core Instruction	Operating systems, files, users, permissions, processes and services.
45 min	Guided Demonstration	Trainer demonstrates user management, permissions, Task Manager/Services and Linux process inspection.
105 min	Hands-on Practice	Learners inspect a Windows/Linux workstation, create users, change permissions and identify unnecessary services.
30 min	Review & Assessment	Short practical check and quiz.

Practical lab / learner output: Workstation security checklist showing user accounts, privilege levels, update status, selected services and file-permission changes.

Assessment: Observed practical task plus 10-question quiz.

DAY 3

Networking Fundamentals for Cybersecurity | Week 1 - Foundations | 5 hours

Learning objectives: By the end of Day 3, learners should be able to:

- Explain how devices communicate across local networks and the internet.
- Identify IP addresses, MAC addresses, ports and common network protocols.
- Use basic network troubleshooting commands.
- Read a simple network diagram and identify security-relevant components.

Core content

- LAN, WAN, internet, routers, switches, access points and firewalls.
- IPv4 addressing, private/public addresses, subnet concepts and MAC addresses.
- TCP and UDP; ports and services.
- DNS, DHCP, ARP, HTTP/HTTPS, SSH, SMTP and common protocol use cases.
- Network troubleshooting and basic network visibility.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Recap plus how data moves from one device to another.
75 min	Core Instruction	IP addressing, TCP/UDP, ports, DNS, DHCP and common protocols.
45 min	Guided Demonstration	Use ipconfig/ifconfig/ip, ping, tracert/traceroute, nslookup/dig and netstat/ss.
105 min	Hands-on Practice	Learners map the training network, test connectivity and identify local addresses, gateway, DNS and active connections.
30 min	Review & Assessment	Network terminology quiz and interpretation exercise.

Practical lab / learner output: A labelled small-network diagram and command-output worksheet showing IP, gateway, DNS, active connections and observed services.

Assessment: Practical command check plus quiz.

DAY 4

Linux Command Line for Cybersecurity |

Week 1 - Foundations | 5 hours

Learning objectives: By the end of Day 4, learners should be able to:

- Comfortably navigate a Linux terminal and file system.
- Use common commands to inspect files, processes, users and network information.
- Understand Linux users, groups and permissions.
- Use pipes, redirection and simple text searching for security tasks.

Core content

- Linux directory structure and shell navigation.
- Commands: pwd, ls, cd, mkdir, cp, mv, rm, cat, less, head, tail and grep.
- Users, groups, sudo, chmod, chown and permission notation.
- Processes and services: ps, top, systemctl and kill.
- Network commands and basic log locations; pipes and redirection.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Linux environment orientation and terminal confidence-building.
75 min	Core Instruction	Files, permissions, users, processes, services and text processing.
45 min	Guided Demonstration	Guided command-line walkthrough using a prepared Ubuntu VM.
105 min	Hands-on Practice	Command challenge: locate files, review logs, change permissions, identify processes and filter text output.
30 min	Review & Assessment	Practical command assessment and recap.

Practical lab / learner output: Completed Linux command challenge sheet with evidence of file, permission, process, service and log tasks.

Assessment: Timed beginner command challenge.

DAY 5

Cyber Threats, Social Engineering & Security Awareness | Week 1 - Foundations | 5 hours

Learning objectives: By the end of Day 5, learners should be able to:

- Recognize major forms of malware and social engineering.
- Identify warning signs in suspicious emails, messages, calls and websites.
- Explain how attackers exploit human behaviour and weak security practices.
- Recommend practical security-awareness controls for an organization.

Core content

- Malware types: viruses, worms, trojans, spyware, ransomware and botnets.
- Phishing, spear phishing, business email compromise, impersonation and social engineering.
- Credential theft, malicious attachments, fake login pages and QR-code scams.
- AI-enabled scams, synthetic content and deepfake-assisted impersonation as awareness risks.
- Security-awareness culture, reporting channels and verification habits.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Threat recap and malware taxonomy.
75 min	Core Instruction	Social engineering methods, phishing indicators and attacker psychology.
45 min	Guided Demonstration	Analyze sample messages, domains, attachments and fake login pages in a safe demonstration.
105 min	Hands-on Practice	Teams classify realistic scenarios, identify indicators and draft a security-awareness message for staff.
30 min	Review & Assessment	Week 1 challenge and quiz.

Practical lab / learner output: Phishing-analysis worksheet and a one-page employee security-awareness advisory.
Assessment: Week 1 challenge: identify and explain the risks in five simulated attack scenarios.

DAY 6

Identity, Passwords & Access Control |

Week 2 - Protection & Defense | 5 hours

Learning objectives: By the end of Day 6, learners should be able to:

- Explain identification, authentication, authorization and accounting.
- Apply strong password and multi-factor authentication practices.
- Understand least privilege, role-based access and account lifecycle management.
- Review a basic access-control setup and identify excessive permissions.

Core content

- Identity and access management (IAM) fundamentals.
- Passwords, passphrases, password managers and credential reuse risks.
- Multi-factor authentication and common authentication factors.
- Least privilege, need-to-know, RBAC and privileged accounts.
- Account provisioning, review, disablement and offboarding.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Review of identity-related breaches and authentication basics.
75 min	Core Instruction	IAM, password security, MFA, least privilege and RBAC.
45 min	Guided Demonstration	Configure local users/groups, MFA examples and permission changes in a lab environment.
105 min	Hands-on Practice	Access review exercise for a fictional company: assign roles, remove excessive privileges and document approvals.
30 min	Review & Assessment	Scenario quiz and access-control review.

Practical lab / learner output: Completed access-control matrix for a small organization with justified roles and permissions.

Assessment: Access-control scenario assessment.

DAY 7

Windows Endpoint Security & Hardening

| Week 2 - Protection & Defense | 5 hours

Learning objectives: By the end of Day 7, learners should be able to:

- Identify the major security controls available on a Windows endpoint.
- Configure essential endpoint protections in a controlled environment.
- Explain patching, anti-malware, host firewall and secure configuration.
- Review basic Windows security events and alerts.

Core content

- Windows security architecture at a beginner level.
- Updates and patch management; Microsoft Defender concepts.
- Windows Firewall, UAC, account protection and device security.
- Secure configuration, application hygiene and removable media risks.
- Introduction to Windows Event Viewer and security logs.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Endpoint threat review and Windows security control overview.
75 min	Core Instruction	Patching, anti-malware, firewall, UAC, secure configuration and logging.
45 min	Guided Demonstration	Trainer hardens a sample Windows VM and demonstrates Defender, Firewall and Event Viewer.
105 min	Hands-on Practice	Learners complete a guided Windows hardening checklist and inspect selected security events.
30 min	Review & Assessment	Practical validation and quiz.

Practical lab / learner output: Windows endpoint hardening checklist and short note on three security events observed.

Assessment: Observed hardening tasks and endpoint-security quiz.

DAY 8

Network Security, Firewalls, Wi-Fi & VPN Fundamentals | Week 2 - Protection & Defense | 5 hours

Learning objectives: By the end of Day 8, learners should be able to:

- Explain the role of firewalls, segmentation, VPNs and secure wireless settings.
- Distinguish trusted and untrusted network zones.
- Design a basic secure network for a small organization.
- Recognize common Wi-Fi and remote-access security weaknesses.

Core content

- Network defense principles and defense in depth.
- Firewalls, allow/deny rules, ingress/egress and host vs network firewalls.
- Network segmentation, guest networks and basic DMZ concepts.
- Wi-Fi security: WPA2/WPA3, strong passphrases, router administration and guest isolation.
- VPN concepts and secure remote access.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Network attack-surface review and defense-in-depth concepts.
75 min	Core Instruction	Firewalls, segmentation, Wi-Fi security and VPN fundamentals.
45 min	Guided Demonstration	Create sample firewall rules and compare insecure vs secure wireless configurations.
105 min	Hands-on Practice	Teams redesign an insecure small-office network and justify zones, firewall rules and Wi-Fi settings.
30 min	Review & Assessment	Design review and quiz.

Practical lab / learner output: A secure small-office network diagram with zones, firewall controls, wireless settings and remote-access recommendations.

Assessment: Network-defense design assessment.

DAY 9

Packet Analysis with Wireshark | Week 2 - Protection & Defense | 5 hours

Learning objectives: By the end of Day 9, learners should be able to:

- Explain what a network packet capture contains.
- Use Wireshark to capture, filter and inspect basic traffic.
- Identify common protocols and normal client-server conversations.
- Spot selected suspicious patterns in prepared traffic samples.

Core content

- Packets, frames, headers and payloads at an introductory level.
- Wireshark interface, capture filters vs display filters.
- DNS, TCP handshakes, HTTP and TLS observations.
- Following streams and interpreting source/destination/ports.
- Basic suspicious-traffic indicators using prepared PCAP files.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Packet-analysis concepts and traffic visibility.
75 min	Core Instruction	Wireshark workflow, filtering and protocol interpretation.
45 min	Guided Demonstration	Live capture demonstration and guided analysis of a prepared PCAP.
105 min	Hands-on Practice	Learners answer investigation questions using PCAPs: identify hosts, domains, protocols, conversations and selected anomalies.
30 min	Review & Assessment	PCAP findings review and quiz.

Practical lab / learner output: Completed packet-analysis worksheet with screenshots or recorded findings from a supplied PCAP.

Assessment: Practical Wireshark investigation.

DAY 10

Introduction to Security Testing & Ethical Hacking | Week 2 - Protection & Defense | 5 hours

Learning objectives: By the end of Day 10, learners should be able to:

- Explain the purpose and boundaries of authorized security testing.
- Describe reconnaissance, scanning, enumeration, validation and reporting at a high level.
- Use basic discovery tools only within an approved lab environment.
- Document findings without attempting harmful exploitation.

Core content

- Ethics, authorization, scope, rules of engagement and safe testing.
- Passive vs active reconnaissance; host and service discovery.
- Introduction to Nmap and interpreting scan results.
- What a vulnerability is and why evidence and context matter.
- Reporting findings, severity and remediation.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Ethics, scope and security-testing lifecycle.
75 min	Core Instruction	Reconnaissance, scanning, enumeration and reporting fundamentals.
45 min	Guided Demonstration	Trainer demonstrates Nmap host/service discovery against an isolated lab target.
105 min	Hands-on Practice	Learners scan authorized lab hosts, identify exposed services and prepare a simple findings table.
30 min	Review & Assessment	Mid-course practical assessment and debrief.

Practical lab / learner output: Authorized lab discovery report listing hosts, services, observations, risk notes and recommended next steps.

Assessment: Mid-course practical assessment covering Days 1-10.

DAY 11

Vulnerability Management & Basic Risk Prioritization | Week 3 - Detection & Application Security | 5 hours

Learning objectives: By the end of Day 11, learners should be able to:

- Explain the vulnerability-management lifecycle.
- Distinguish vulnerabilities, exposures, misconfigurations and patches.
- Interpret basic vulnerability-scan results and severity information.
- Prioritize remediation based on risk, exposure and business impact.

Core content

- Vulnerability discovery, validation, prioritization, remediation and verification.
- CVEs, severity scoring concepts and why severity is not the same as business risk.
- Patch management and configuration weaknesses.
- False positives and the importance of validation.
- Writing clear remediation recommendations.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	From scanning to vulnerability management.
75 min	Core Instruction	Vulnerability identifiers, severity, risk context and remediation lifecycle.
45 min	Guided Demonstration	Review a prepared scanner report and validate selected findings against a lab system.
105 min	Hands-on Practice	Teams triage a vulnerability report, rank issues and produce a remediation plan.
30 min	Review & Assessment	Prioritization review and quiz.

Practical lab / learner output: Prioritized vulnerability register with owner, severity, business context, action and target remediation date.

Assessment: Vulnerability triage exercise.

DAY 12

Web & Application Security Fundamentals | Week 3 - Detection & Application Security | 5 hours

Learning objectives: By the end of Day 12, learners should be able to:

- Explain how a browser communicates with a web application.
- Understand requests, responses, headers, cookies, sessions and input validation.
- Recognize major categories of web application security weaknesses.
- Use browser developer tools and an intercepting proxy in a safe lab.

Core content

- Client-server model, URLs, HTTP methods, status codes and HTTPS.
- Cookies, sessions, authentication and authorization in web applications.
- Input validation and common weakness categories such as injection and broken access control.
- OWASP-style application security concepts at an introductory level.
- Browser developer tools and Burp Suite Community basics.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	How web applications work and where security controls sit.
75 min	Core Instruction	HTTP, sessions, authentication, authorization and common web weaknesses.
45 min	Guided Demonstration	Inspect requests in browser developer tools and Burp against an intentionally vulnerable training app.
105 min	Hands-on Practice	Guided lab: map application requests, identify selected weaknesses and explain the security impact without attacking external systems.
30 min	Review & Assessment	Web-security quiz and findings discussion.

Practical lab / learner output: Basic web-application observation report showing requests, session behaviour, identified weakness categories and remediation ideas.

Assessment: Guided web-security lab assessment.

DAY 13

Cryptography, Encryption & Data Protection | Week 3 - Detection & Application Security | 5 hours

Learning objectives: By the end of Day 13, learners should be able to:

- Distinguish encryption, hashing and encoding.
- Explain symmetric and asymmetric cryptography at a beginner level.
- Understand certificates, TLS/HTTPS and digital signatures conceptually.
- Apply basic data-protection practices to files and communications.

Core content

- Why cryptography matters; plaintext, ciphertext and keys.
- Symmetric vs asymmetric encryption.
- Hashing, integrity checks, password hashing and salting concepts.
- PKI, certificates, HTTPS/TLS and digital signatures.
- Data at rest, data in transit, backups and secure disposal.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Data-protection goals and cryptography vocabulary.
75 min	Core Instruction	Encryption, hashing, keys, certificates, TLS and signatures.
45 min	Guided Demonstration	Hash sample files, compare integrity, inspect a website certificate and demonstrate simple file encryption.
105 min	Hands-on Practice	Learners perform integrity checks, inspect certificates and develop a data-protection plan for sample information types.
30 min	Review & Assessment	Concept quiz and practical verification.

Practical lab / learner output: Data-classification and protection worksheet plus file-hash evidence and certificate observations.

Assessment: Cryptography concepts quiz and lab check.

DAY 14

Cloud, Email & Online Service Security |

Week 3 - Detection & Application Security | 5 hours

Learning objectives: By the end of Day 14, learners should be able to:

- Explain shared responsibility in cloud services at a beginner level.
- Identify common security risks in cloud storage, email and SaaS accounts.
- Apply account, sharing and access controls to online services.
- Recognize risky configurations such as public links and excessive permissions.

Core content

- Cloud service models and shared-responsibility concept.
- SaaS account security, MFA, session management and recovery options.
- Cloud storage permissions, public sharing and link controls.
- Email account security, forwarding rules and suspicious login activity.
- Backups, account recovery and secure collaboration.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Cloud and SaaS security concepts.
75 min	Core Instruction	Shared responsibility, account protection, permissions and secure collaboration.
45 min	Guided Demonstration	Review sample cloud-storage and email security settings; demonstrate risky vs safer configurations.
105 min	Hands-on Practice	Learners audit a fictional organization's online-service settings and produce corrective actions.
30 min	Review & Assessment	Cloud-security scenario quiz.

Practical lab / learner output: Online-services security checklist covering identity, sharing, recovery, backup and monitoring settings.

Assessment: Configuration-review scenario.

DAY 15

Security Monitoring, Logs & SOC Fundamentals | Week 3 - Detection & Application Security | 5 hours

Learning objectives: By the end of Day 15, learners should be able to:

- Explain why security logs and monitoring are important.
- Identify useful log sources from endpoints, networks and applications.
- Understand alerts, indicators of compromise and basic triage.
- Perform a simple investigation using prepared log data.

Core content

- Logging fundamentals: timestamp, event, user, source, destination and action.
- Windows, Linux, firewall, authentication and web logs.
- SIEM concepts and the purpose of centralized monitoring.
- Indicators of compromise (IOCs), baselines and suspicious patterns.
- Alert triage: what happened, when, where, who and what next.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Detection mindset and log fundamentals.
75 min	Core Instruction	Log sources, SIEM concepts, IOCs and alert triage.
45 min	Guided Demonstration	Trainer walks through prepared endpoint/authentication logs and a simple SIEM-style view.
105 min	Hands-on Practice	Learners investigate a supplied log set, build an event timeline and identify suspicious activity.
30 min	Review & Assessment	Week 3 challenge and review.

Practical lab / learner output: Incident timeline and analyst notes based on supplied log evidence.

Assessment: Week 3 log-analysis challenge.

DAY 16

Incident Response & Containment | Week 4 - Response, Governance & Career Readiness | 5 hours

Learning objectives: By the end of Day 16, learners should be able to:

- Describe the stages of an incident-response process.
- Classify and escalate common security incidents.
- Recommend safe containment and recovery actions.
- Document an incident clearly and preserve relevant evidence.

Core content

- Preparation, detection/analysis, containment, eradication, recovery and lessons learned.
- Incident severity, roles, escalation and communications.
- Containment choices for compromised accounts, endpoints and network activity.
- Evidence preservation and maintaining an incident record.
- Post-incident review and control improvement.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	What makes an event an incident and how response is organized.
75 min	Core Instruction	Incident-response lifecycle, severity, escalation, containment and recovery.
45 min	Guided Demonstration	Facilitator runs a tabletop scenario involving a compromised employee account and suspicious endpoint.
105 min	Hands-on Practice	Teams respond to timed injects, make containment decisions and keep an incident log.
30 min	Review & Assessment	Tabletop debrief and response quiz.

Practical lab / learner output: Completed incident record containing timeline, decisions, containment actions, communications and recovery steps.

Assessment: Team incident-response tabletop assessment.

DAY 17

Digital Forensics & Evidence Fundamentals | Week 4 - Response, Governance & Career Readiness | 5 hours

Learning objectives: By the end of Day 17, learners should be able to:

- Explain the purpose of digital forensics and the importance of preserving evidence.
- Identify common sources of digital evidence.
- Understand metadata, timestamps, file hashes and chain of custody.
- Perform a basic review of prepared evidence without modifying the source.

Core content

- Digital forensics purpose, principles and limitations.
- Volatile vs non-volatile evidence; endpoint, network and account evidence.
- Hashes, integrity and chain of custody.
- File metadata, timestamps, browser history and basic artifacts.
- Forensic notes and evidence handling.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Forensics principles and evidence preservation.
75 min	Core Instruction	Evidence sources, hashes, metadata, timestamps and chain of custody.
45 min	Guided Demonstration	Trainer demonstrates hashing, metadata review and read-only examination of prepared evidence.
105 min	Hands-on Practice	Learners examine supplied files/logs, record artifacts and build a short evidence summary.
30 min	Review & Assessment	Evidence-handling check and quiz.

Practical lab / learner output: Basic evidence worksheet including hashes, metadata, notable artifacts and chain-of-custody entries.

Assessment: Forensics handling and interpretation exercise.

DAY 18

Cybersecurity Governance, Risk, Privacy & Policies

Week 4 - Response, Governance & Career Readiness | 5 hours

Learning objectives: By the end of Day 18, learners should be able to:

- Explain the role of governance, policies, standards and procedures in cybersecurity.
- Use a simple likelihood-impact approach to discuss cyber risk.
- Understand privacy, data handling and regulatory compliance as organizational responsibilities.
- Draft practical security policies for a small organization.

Core content

- Governance, accountability and security roles.
- Risk identification, likelihood, impact, risk treatment and residual risk.
- Policies, standards, procedures and guidelines.
- Privacy and data-protection principles: lawful handling, minimization, access control, retention and breach response.
- Third-party risk, acceptable use, password, backup and incident-response policies.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Governance and risk concepts for non-specialists.
75 min	Core Instruction	Risk treatment, policy structure, privacy principles and third-party security.
45 min	Guided Demonstration	Review examples of strong and weak policy statements and a simple risk register.
105 min	Hands-on Practice	Teams create a mini risk register and draft one short security policy for a fictional organization.
30 min	Review & Assessment	Policy peer review and quiz.

Practical lab / learner output: Five-item cyber risk register and a concise organizational security policy.

Assessment: Risk-and-policy assessment.

DAY 19

Security Operations, Career Pathways & Capstone Preparation | Week 4 - Response, Governance & Career Readiness | 5 hours

Learning objectives: By the end of Day 19, learners should be able to:

- Connect the course skills to entry-level cybersecurity job functions.
- Explain common responsibilities in SOC, support, GRC, vulnerability management and security operations roles.
- Communicate technical findings clearly to non-technical stakeholders.
- Prepare a structured investigation plan for the final capstone.

Core content

- Entry-level cybersecurity roles and common work activities.
- How tickets, alerts, evidence, escalation and reporting flow in security teams.
- Technical documentation: issue, evidence, impact, action and owner.
- Professional ethics, continuous learning and portfolio-building.
- Capstone briefing, team roles, scoring rubric and lab readiness.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Map learned skills to real cybersecurity work.
75 min	Core Instruction	SOC workflow, documentation, communication and career pathways.
45 min	Guided Demonstration	Trainer demonstrates how to turn raw findings into a concise analyst report and stakeholder briefing.
105 min	Hands-on Practice	Learners complete a mini triage exercise, improve their CV/skills map and plan capstone roles.
30 min	Review & Assessment	Capstone readiness check and practical review.

Practical lab / learner output: Personal cybersecurity skills map plus team capstone investigation plan.

Assessment: Capstone readiness assessment.

DAY 20

Integrated Cybersecurity Capstone & Final Assessment | Week 4 - Response, Governance & Career Readiness | 5 hours

Learning objectives: By the end of Day 20, learners should be able to:

- Apply multiple cybersecurity skills to a realistic simulated incident.
- Collect and interpret endpoint, network, log and application evidence.
- Prioritize risks and recommend containment and remediation actions.
- Produce and present a concise professional incident report.

Core content

- Integrated scenario involving suspicious email activity, compromised credentials, endpoint alerts, network evidence and a vulnerable web service.
- Incident triage, evidence review, packet/log analysis and vulnerability prioritization.
- Containment, recovery and prevention recommendations.
- Technical reporting and executive communication.
- Final knowledge review and course reflection.

Five-hour delivery plan

Time	Learning Block	Focus / Activity
45 min	Foundation	Capstone briefing, team roles, evidence pack and rules.
75 min	Core Instruction	Initial triage: identify likely attack path, affected assets and investigation priorities.
45 min	Guided Demonstration	Facilitator checkpoint: teams validate their evidence and investigation approach.
105 min	Hands-on Practice	Full team investigation, response plan, report preparation and short presentation.
30 min	Review & Assessment	Final presentations, individual quiz, feedback and course close.

Practical lab / learner output: Team incident report and 5-7 minute presentation covering what happened, evidence, affected assets, risk, containment, remediation and lessons learned.

Assessment: Final capstone (team) plus individual end-of-course knowledge assessment.

4. Assessment & Certification Framework

A balanced approach that rewards practical performance

Assessment Component	Weight	Purpose
Daily practical labs and worksheets	**35%**	Measures the learner's ability to apply daily skills and document findings.
Daily/weekly knowledge quizzes	**10%**	Checks understanding of key terminology, concepts and decisions.
Week 1 and Week 3 practical challenges	**15%**	Assesses synthesis of multiple skills at key points in the programme.
Mid-course practical assessment (Day 10)	**15%**	Validates foundational operating system, networking and security-testing skills.
Final capstone and presentation (Day 20)	**25%**	Assesses integrated investigation, response, reporting and communication skills.

Recommended certification requirements

- Minimum 80% attendance.
- Minimum 70% overall assessment score.
- Completion of the Day 20 capstone and final presentation.
- Compliance with ethical-use, lab-safety and authorized-testing rules.

The implementing organization may adjust the pass mark or attendance requirement to meet sponsor, institutional or regulatory conditions.

5. Lab Environment, Tools & Training Resources

Minimum setup for practical delivery

Category	Recommended Requirement
Learner workstation	Laptop/desktop capable of running a browser plus at least one virtual machine; 8 GB RAM minimum recommended, 16 GB preferred for smoother labs.
Operating systems	Windows 10/11 where available and Ubuntu Linux in a virtual machine or prepared lab environment.
Virtualization	VirtualBox, VMware Workstation Player/Pro, or an institution-provided virtual lab platform.
Networking tools	Built-in Windows/Linux networking commands, Wireshark and Nmap.
Web security lab	Browser developer tools, Burp Suite Community Edition and an intentionally vulnerable training application such as OWASP Juice Shop or equivalent local lab.
Monitoring/logs	Windows Event Viewer, Linux logs and prepared log datasets; optional Wazuh or another instructor-provided SIEM environment.
Learning materials	Trainer slides, learner workbook, lab guides, sample PCAPs, sample logs, policy templates, quizzes, capstone evidence pack and assessment rubrics.
Connectivity	Reliable local network and internet access for updates/reference materials; core labs should be prepared so learning can continue if internet access is unstable.
Safety	All scanning, interception and security-testing exercises must remain within approved training systems, isolated virtual labs or explicitly authorized targets.

Trainer profile

- Practical cybersecurity experience across operating systems, networking and security operations.
- Ability to explain technical concepts to first-time learners without assuming prior knowledge.
- Comfortable facilitating Windows/Linux labs, Wireshark, Nmap, web-security training labs and incident-response scenarios.
- Strong emphasis on ethics, authorization, learner safety, documentation and defensive security practice.

6. Final Capstone Design

Integrated incident simulation for Day 20

Scenario: A fictional organization reports suspicious emails, an employee account showing unusual activity, an endpoint generating alerts, unexplained network connections and a customer-facing application with security concerns. Teams receive a controlled evidence pack and must investigate the incident without using any external target.

Required team tasks

1. Triage the incident and identify the most important questions to answer.
2. Analyze supplied phishing/email indicators, endpoint evidence, network traffic and logs.
3. Identify affected assets, likely attack path and relevant vulnerabilities or misconfigurations.
4. Recommend immediate containment and safe recovery actions.
5. Prioritize remediation and prevention controls.
6. Prepare a concise incident report with evidence, impact, actions and lessons learned.
7. Present findings to a simulated management audience in 5-7 minutes.

Capstone scoring rubric

Dimension	Weight	What good performance looks like
Investigation & evidence	**30%**	Findings are supported by the supplied evidence and the team follows a logical investigation path.
Technical accuracy	**25%**	Threats, vulnerabilities, controls and recommendations are correctly understood.
Response & remediation	**20%**	Containment, recovery and prevention actions are prioritized and practical.
Report quality	**15%**	Report is concise, structured, evidence-based and understandable.
Presentation & teamwork	**10%**	Team communicates clearly, shares roles effectively and answers questions confidently.

END OF CURRICULUM